

CSS @page At-Rule

Complete Guide: From Basics to Advanced

CSS Guide

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Chapter 1

Introduction

1.1 What is the @page At-Rule?

The `@page` at-rule is a CSS feature that defines properties of a printed page, such as size, margins, page orientation, and margin boxes for headers and footers. It's used exclusively in print media to control how web content appears when printed or viewed in print preview.

```
/* Basic syntax */
@page {
    size: A4;
    margin: 1in;
}

/* With pseudo-class */
@page :first {
    margin-top: 2in;
}
```

The `@page` rule revolutionizes print design by enabling CSS-only control over page layout, dimensions, and content placement without requiring JavaScript or separate print templates.

1.2 Why @page Matters

1.2.1 Real-World Applications

- **Professional Documents:** Invoices, reports, contracts formatted for printing
- **Publishing:** Books, magazines, multi-page documents with consistent styling
- **Certificates:** Official documents with specific dimensions and layouts
- **Forms:** Government or business forms with precise margins and page breaks
- **Tickets:** Event tickets, boarding passes, receipts with exact sizing
- **Legal Documents:** Contracts requiring specific page formatting standards

1.2.2 Benefits

- Control exact page dimensions for printing
- Set margins and margins boxes (headers/footers)
- Define page orientation (portrait/landscape)
- Create pseudo-pages for first, left, right pages
- Professional document presentation
- CSS-only solution without JavaScript

1.3 Browser Support

The @page rule has varying levels of support across browsers:

Browser	Basic Support	Margin Boxes	Pseudo-classes
Chrome	Full	Partial	Full
Firefox	Full	Limited	Full
Safari	Limited		Limited
Edge	Full	Partial	Full

Chapter 2

Core Concepts

2.1 @page Syntax and Structure

2.1.1 Basic Syntax

```
/* Simple @page rule */
@page {
    size: A4;
    margin: 1in;
    background: white;
}

/* With named page */
@page mypage {
    size: letter;
    margin: 0.5in;
}

/* With pseudo-class */
@page :first {
    margin-top: 2in;
}

/* Using named page */
.chapter {
    page: mypage;
}
```

2.2 Page Size Properties

2.2.1 Predefined Page Sizes

Size	Dimensions	Common Use
letter	8.5" × 11"	US standard
legal	8.5" × 14"	Legal documents
A4	210mm × 297mm	International standard
A3	297mm × 420mm	Larger format
A5	148mm × 210mm	Smaller format
B4	257mm × 364mm	B series
B5	176mm × 250mm	B series small
C5	162mm × 229mm	Envelope
tabloid	11" × 17"	Newspaper
ledger	17" × 11"	Wide format

2.2.2 Page Size Examples

```
/* Predefined size */
@page {
    size: A4;
}

/* Size with orientation */
@page {
    size: letter landscape;
}

@page {
    size: A4 portrait;
}

/* Custom size */
@page {
    size: 100mm 150mm;
}

/* Width and height */
@page {
    size: 8.5in 11in;
}

/* Auto orientation */
@page {
    size: auto;
}
```

2.3 Page Margin Properties

2.3.1 Margin Setting

```
/* All margins equal */
@page {
    margin: 1in;
}

/* Top/bottom, left/right */
@page {
    margin: 1in 0.5in;
}

/* Top, right/left, bottom */
@page {
    margin: 1.5in 1in 1in;
}

/* Individual margins */
@page {
    margin-top: 1in;
    margin-right: 0.75in;
    margin-bottom: 1in;
    margin-left: 0.75in;
}

/* Using margin shorthand */
@page {
    margin: 1in;
}
```

2.3.2 Margin Box Properties

```
/* Define content in margin boxes */
@page {
    @top-left {
        content: "Header Left";
    }

    @top-center {
        content: "Header Center";
    }

    @top-right {
        content: "Header Right";
    }

    @bottom-left {
        content: "Footer Left";
    }
}
```

```
@bottom-center {
    content: "Page " counter(page);
}

@bottom-right {
    content: "Footer Right";
}
}
```

2.4 @page Pseudo-classes

2.4.1 Available Pseudo-classes

- **:first** - First page
- **:left** - Left pages (for double-sided printing)
- **:right** - Right pages (for double-sided printing)
- **:blank** - Blank pages

2.4.2 Pseudo-class Examples

```
/* First page styling */
@page :first {
    margin-top: 2in;
    size: letter;
}

/* Left pages (even pages) */
@page :left {
    margin-left: 1.5in;
    margin-right: 0.75in;
}

/* Right pages (odd pages) */
@page :right {
    margin-left: 0.75in;
    margin-right: 1.5in;
}

/* Left page headers */
@page :left {
    @bottom-left {
        content: counter(page);
    }
}

/* Right page headers */
@page :right {
    @bottom-right {
        content: counter(page);
    }
}
```

```
}  
}
```

Chapter 3

Practical Examples

3.1 Example 1: Professional Document

```
/* Standard 8.5" x 11" document */
@page {
    size: letter;
    margin: 1in;
}

/* Title page: larger top margin */
@page :first {
    margin-top: 2in;
}

/* Headers and footers */
@page {
    @top-center {
        content: "Company Name";
        font-weight: bold;
    }

    @bottom-center {
        content: "Page " counter(page) " of " counter(pages);
        font-size: 10pt;
    }

    @bottom-right {
        content: string(date);
        font-size: 9pt;
    }
}

/* Named pages for different sections */
@page chapter {
    margin: 1.25in;
}

@page appendix {
    margin: 1in;
}
```

```
        background: linear-gradient(to right,
                                    white 95%,
                                    #f0f0f0 100%);
    }

    /* Chapter breaks on new page */
    .chapter {
        page: chapter;
        page-break-before: always;
    }

    /* Appendix section */
    .appendix {
        page: appendix;
    }
```

3.2 Example 2: Book Publishing

```
/* A4 page for book */
@page {
    size: A4;
    margin: 1.5in 1in;
}

/* Title page: centered, more space at top */
@page :first {
    margin-top: 3in;
    margin-bottom: 3in;
}

/* Left pages (even pages) */
@page :left {
    margin-left: 1.5in;
    margin-right: 1in;

    @bottom-left {
        content: counter(page);
    }

    @bottom-right {
        content: string(chapter-title);
    }
}

/* Right pages (odd pages) */
@page :right {
    margin-left: 1in;
    margin-right: 1.5in;

    @bottom-left {
        content: string(section-title);
    }
}
```

```
    @bottom-right {
        content: counter(page);
    }
}

/* Running headers */
h1 {
    string-set: chapter-title content();
}

h2 {
    string-set: section-title content();
}
```

3.3 Example 3: Multi-Section Report

```
/* Executive Summary: different size */
@page executive-summary {
    size: letter;
    margin: 1.5in;
}

/* Main content: standard letter */
@page main-content {
    size: letter;
    margin: 1in;

    @top-left {
        content: "Confidential Report";
        color: red;
    }
}

/* Appendix: compact margins */
@page appendix-page {
    size: letter;
    margin: 0.75in;

    @bottom-center {
        content: "Appendix - Page " counter(page);
    }
}

/* Section assignment */
.executive-summary {
    page: executive-summary;
}

.main-content {
    page: main-content;
}
```

```
.appendix {
    page: appendix-page;
}

/* Page break control */
.new-section {
    page-break-before: always;
}
```

3.4 Example 4: Invoices and Receipts

```
/* Invoice layout */
@page invoice {
    size: letter;
    margin: 0.5in;
}

/* Each invoice on new page */
.invoice {
    page: invoice;
    page-break-after: always;
}

/* Last invoice: no break after */
.invoice:last-of-type {
    page-break-after: auto;
}

/* Invoice header with no footer on first page */
@page invoice :first {
    @bottom-center {
        content: "";
    }
}

/* Receipt (narrow format) */
@page receipt {
    size: 100mm auto;
    margin: 5mm;
}

.receipt {
    page: receipt;
    page-break-after: always;
}

/* Receipt footer */
@page receipt {
    @bottom-center {
        content: "Thank you for your purchase";
        font-size: 8pt;
    }
}
```

```
}  
}
```

3.5 Example 5: Ticket Printing

```
/* Ticket dimensions: 100mm x 150mm */  
@page ticket {  
    size: 100mm 150mm;  
    margin: 0;  
    background: white;  
}  
  
/* Multiple tickets per page */  
.ticket {  
    page: ticket;  
    width: 100mm;  
    height: 150mm;  
    box-sizing: border-box;  
    page-break-after: always;  
}  
  
/* Last ticket: no page break */  
.ticket:last-of-type {  
    page-break-after: auto;  
}  
  
/* Ticket content styling */  
@page ticket {  
    /* No headers/footers for tickets */  
    margin: 0;  
    padding: 0;  
}  
  
/* Barcode positioning */  
.barcode {  
    position: absolute;  
    bottom: 10mm;  
    right: 10mm;  
}  
  
/* Tear line */  
.tear-line {  
    border-top: 2px dashed #ccc;  
    margin: 5mm 0;  
}
```

Chapter 4

Advanced Techniques

4.1 Page Counters and Strings

```
/* Reset counter for new section */
.new-chapter {
    counter-reset: page 1;
}

/* Display total pages */
@page {
    @bottom-center {
        content: "Page " counter(page) " of " counter(pages);
    }
}

/* Using string() for running headers */
h1 {
    string-set: title content();
}

@page {
    @top-center {
        content: string(title);
    }
}
```

4.2 Conditional Page Styling

```
/* First page special styling */
@page :first {
    background: linear-gradient(to bottom,
                                #667eea 0%,
                                white 100%);
    margin-top: 2in;
}

/* Blank pages styling */
```

```
@page :blank {
    margin: 0;
    display: none;
}

/* Alternating margins for double-sided */
@page :left {
    margin-left: 1.5in;
}

@page :right {
    margin-right: 1.5in;
}
```

4.3 Combining with Print Media

```
@media print {
    @page {
        size: A4;
        margin: 1in;
    }

    @page :first {
        margin-top: 2in;
    }

    /* Remove screen elements */
    nav, .sidebar, .ads {
        display: none;
    }

    /* Adjust content for print */
    body {
        font-size: 12pt;
    }
}
```

Chapter 5

Browser Support & Limitations

5.1 Feature Support Matrix

Feature	Chrome	Firefox	Safari	Edge
Page size	Yes	Yes	Limited	Yes
Margins	Yes	Yes	Yes	Yes
Margin boxes	Partial	Limited		Partial
Pseudo-classes	Yes	Yes	Limited	Yes
Named pages	Yes	Yes	Limited	Yes

5.2 Known Limitations

- **Margin boxes:** Limited support for complex content
- **Safari:** Limited @page support overall
- **Cross-browser:** Margin box support varies significantly
- **Counters:** Some limitations with reset and scope
- **Named pages:** Browser support varies

5.3 Progressive Enhancement

```
/* Fallback: basic styling that works everywhere */
@page {
    margin: 1in;
}

/* Enhanced: fancy styling for supported browsers */
@supports (page-break-inside: avoid) {
    @page {
        size: A4;
        margin: 1.5in 1in;
    }

    @page :first {
```

```
    margin-top: 2in;  
  }  
}
```

Chapter 6

Best Practices

6.1 Key Recommendations

6.1.1 1. Test Across Browsers

Always test @page styling in:

- Chrome/Edge print preview
- Firefox print preview
- Actual printer output
- Different page sizes

6.1.2 2. Use Standard Page Sizes

```
/* Good: Standard size */
@page {
    size: A4;
    margin: 1in;
}

/* Avoid: Unusual sizes can cause issues */
@page {
    size: 200mm 300mm;
}
```

6.1.3 3. Set Sensible Margins

```
/* Good: Readable margins */
@page {
    margin: 1in;
}

/* Avoid: Too small margins */
@page {
    margin: 0.1in;
}
```

6.1.4 4. Combine with Page Break Control

```
@media print {  
  @page {  
    size: A4;  
    margin: 1in;  
  }  
  
  /* Control breaks */  
  h2 {  
    page-break-after: avoid;  
  }  
  
  table {  
    page-break-inside: avoid;  
  }  
}
```

6.2 Common Patterns

Pattern	Use Case
Standard letter/A4	Most documents
Large top margin	Cover pages
Different left/right	Double-sided printing
Multiple named pages	Multi-section documents
Header/footer text	Professional documents

Chapter 7

Common Issues & Solutions

7.1 Issue 1: Margin Boxes Not Appearing

Problem: Header/footer content not printing.

Causes:

- Browser doesn't support margin boxes
- Content property not set correctly
- Page margin too small

Solution:

```
@page {
  size: letter;
  margin: 1in;

  /* Ensure margin boxes are defined */
  @bottom-center {
    content: "Page " counter(page);
    font-size: 10pt;
  }
}

/* Use fallback styling */
.footer {
  display: none;
}

@media print {
  .footer {
    position: fixed;
    bottom: 1in;
    width: 100%;
    text-align: center;
  }
}
```

7.2 Issue 2: Page Size Not Recognized

Problem: Custom page size not working.

Solution:

```
/* Use standard sizes */
@page {
    size: A4; /* Works everywhere */
}

/* Avoid custom unless necessary */
@page {
    size: 100mm 150mm; /* Limited support */
}

/* With fallback */
@page {
    size: A4;
    margin: 1in;
}
```

7.3 Issue 3: Content Overflowing Pages

Problem: Content extends beyond page margins.

Solution:

```
@page {
    size: letter;
    margin: 1in;
}

/* Adjust content width */
body {
    max-width: 6.5in; /* letter - margins */
    box-sizing: border-box;
}

/* Use page break control */
@media print {
    h2 {
        page-break-after: avoid;
    }

    table {
        page-break-inside: avoid;
    }
}
```

Chapter 8

Summary & Quick Reference

8.1 When to Use @page

Use @page when:

- Creating professional printed documents
- Need specific page dimensions
- Require custom margins
- Building multi-page documents
- Printing invoices or receipts
- Creating certificates or tickets

8.2 Quick Reference

8.2.1 Basic @page Template

```
/* Standard document */
@page {
    size: letter;
    margin: 1in;
}

/* First page styling */
@page :first {
    margin-top: 1.5in;
}

/* Headers and footers */
@page {
    @top-center {
        content: "Document Title";
    }

    @bottom-center {
```

```
        content: "Page " counter(page);
    }
}

/* Apply in print media */
@media print {
    /* Additional print styles */
}
```

8.2.2 Common Page Sizes Reference

```
/* Letter: 8.5" x 11" (US) */
@page { size: letter; }

/* A4: 210mm x 297mm (International) */
@page { size: A4; }

/* Legal: 8.5" x 14" */
@page { size: legal; }

/* A3: 297mm x 420mm */
@page { size: A3; }

/* Custom width x height */
@page { size: 100mm 150mm; }

/* Landscape orientation */
@page { size: letter landscape; }

/* Portrait orientation */
@page { size: A4 portrait; }
```

8.2.3 Complete Document Template

```
@media print {
    @page {
        size: letter;
        margin: 1in;
    }

    @page :first {
        margin-top: 2in;
    }

    @page {
        @top-center {
            content: "Company Name";
            font-weight: bold;
        }

        @bottom-center {
```

```
        content: "Page " counter(page)
                  " of " counter(pages);
    }
}

/* Remove screen elements */
nav, .sidebar, button {
    display: none;
}

/* Optimize for print */
body {
    font-size: 12pt;
    color: black;
    background: white;
}

h2 {
    page-break-after: avoid;
}

table {
    page-break-inside: avoid;
}
}
```

Thank you for reading!

For more information, visit MDN Web Docs and web.dev